These approaches to imputing missing Q are likely to be somewhat conservative and result in lower values for Q than the actual limit of quantification when the nutrient's effect is positive and larger values of Q than the actual limit when the nutrient's effect is negative.

We can summarize the decision rules for imputation that result from the above assumptions as follows:

Case 1: C, D and Q are known	**Impute Trace As:**	**Impute Not Detected As:**
When *More* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	$\frac{C + D}{2}$
When *Less* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	$\frac{C + D}{2}$

Case 2: D & Q known, C unknown	**Impute Trace As:**	**Impute Not Detected As:**
When *More* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	$\frac{0 + D}{2}$
When *Less* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	D

Case 3: C & Q known, D unknown	**Impute Trace As:**	**Impute Not Detected As:**
When *More* is Better:	$\sqrt{\frac{C^2 + Q^2}{2}}$	C
When *Less* is Better:	Q	$\frac{C + Q}{2}$

Case 4: Q known, C & D unknown	**Impute Trace As:**	**Impute Not Detected As:**
When *More* is Better:	$\sqrt{\frac{0^2 + Q^2}{2}}$	0
When *Less* is Better:	Q	Q

Case 5: C & D known, Q unknown	**Impute Trace As:**	**Impute Not Detected As:**
When *More* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	$\frac{C + D}{2}$
When *Less* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	$\frac{C + D}{2}$

Case 6: D known, Q &C unknown	Impute Trace As:	Impute Not Detected As:
When *More* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	$\frac{0 + D}{2}$
When *Less* is Better:	$\sqrt{\frac{D^2 + Q^2}{2}}$	D

Case 7: C known, D & Q unknown	Impute Trace As:	Impute Not Detected As:
When *More* is Better:	$\sqrt{\frac{C^2 + Q^2}{2}}$	C
When *Less* is Better:	Q	$\frac{C + Q}{2}$

Case 8: D, Q & C unknown	Impute Trace As:	Impute Not Detected As:
When *More* is Better:	$\sqrt{\frac{0^2 + Q^2}{2}}$	0
When *Less* is Better:	Q	Q

However, there are some nutrients that have positive dietary effects when they are present in a certain range of quantities and have negative dietary effects when they are present in levels outside that range. For such nutrients the following procedure is recommended:

1. If other available data for the nutrient are towards the minimum of the beneficial

range, follow the procedures for imputing for “More is Better” nutrients.

2. If other available data for the nutrient are towards the maximum of the beneficial range, follow the procedures for imputing for “Less is Better” nutrients.

3. If other available data for the nutrient are distributed throughout the beneficial range impute twice, once with the “More is Better” method, and another time with the “Less is Better” method. Use the average of the two values for imputation.

References for Appendix E

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